Dylan Dsouza

San Diego, CA

dydsouza@ucsd.edu | dylandsouza.com | linkedin.com/in/dsouza-dylan | github.com/dsouza-dylan

Education

University of California San Diego — San Diego, CA
Bachelor of Science in Data Science – GPA: 3.92 (Provost Honors) — *Expected March 2027*

Experience

Data Engineer Intern — Jul 2026 – Present
Capgemini — *Mumbai, India*
- Building a serverless AWS RAG pipeline for natural language Q&A over PDFs: chunking documents, generating embeddings via Amazon Bedrock, indexing in FAISS, and parallelizing chunk retrieval across AWS Lambda for cost-efficient, low-latency responses.
- Underwent structured generative AI training covering prompt engineering frameworks, MCP protocol, agentic AI workflow design, RAG and vector database architectures, and AI governance and security.

AI Engineer — Oct 2025 – Jun 2026
Computer Science & Engineering Society — *San Diego, CA*
- Implemented a multi-stage Neo4j retrieval pipeline fusing MiniLM vector search and BM25 via RRF, MMR diversity re-ranking, and cross-encoder reranking, and built the React/Vite frontend for an agentic AI platform modeling professional decision-making styles.
- Constructed a knowledge graph pipeline extracting LLM-generated concepts from research documents, mapping entity relationships via contextual proximity scoring and Girvan-Newman community detection across 8+ entity categories.

Machine Learning Engineer — Oct 2025 – May 2026
Engineers for Exploration — *San Diego, CA*
- Engineered a Google Earth Engine pipeline staging Sentinel-2 optical bands fused with 64-band AlphaEarth satellite embeddings across training tiles in Madagascar and Mozambique, gating downloads to mangrove regions via ESA WorldCover classification.
- Fine-tuned a continual learning segmentation model on 21,000+ GeoTIFF image-mask pairs, achieving 82–85% mIoU across 4 land-cover classes for global-scale pixel-level mangrove mapping.

Data Analyst — May 2025 – May 2026
Scripps Institution of Oceanography — *San Diego, CA*
- Automated 11 years of coral reef data processing (60% cut in manual processing time), building a Python ETL pipeline that standardized 40,000+ benthic annotations into clean datasets for downstream analysis across 15 reef sites.
- Conducted EDA and time series analyses across 5 experimental groups and 16 survey time points to investigate treatment effects on reef dynamics, identifying temporal trends and ecological patterns to inform restoration strategies.

Market Research Intern — Jul 2023 – Aug 2023
Bennett Coleman & Co. Ltd. (The Times Group) — *Mumbai, India*
- Built a multiclass NLP classifier achieving 85% accuracy across 5 market sectors, processing 56,000+ newspaper headlines using TF-IDF and BERT embeddings to automate market sector categorization at scale.
- Produced an internal research brief on data infrastructure modernization – covering data lakes, CDPs, and CRM optimization – to support a 5-person team's evaluation of editorial and business workflow improvements.

Projects

ENSOcast: Climate Analytics Dashboard | *Conference Project (SDUTC 2025)* — View Project | GitHub
- Engineered a Streamlit climate dashboard on 40+ years of NOAA data; benchmarked Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble models achieving 82% accuracy in 3-class ENSO phase prediction.

Who Gets the Old Planes? | *Personal Project* — View Project | GitHub
- Cross-referenced 2.7M flights with FAA aircraft registration records to map narrowbody fleet age across U.S. airports, revealing how geography and market dynamics shape the age of aircraft passengers board.

DegreeDelta: Quantifying Education Returns | *Course Project (Statistical Methods)* — View Project | GitHub
- Quantified the college income premium across all 50 U.S. states using weighted ACS PUMS microdata and state-level aggregation, revealing geographic variation in the financial return to higher education.

America on Fire: Mapping U.S. Wildfire Distribution | *Course Project (Data Visualization)* — View Project | GitHub
- Visualized 2024 U.S. wildfire distribution through an interactive D3.js map using NASA MODIS satellite data, surfacing spatial burn patterns, seasonal trends, and unexpected regional hotspots.

Predicting U.S. Power Outage Severity | *Personal Project* — View Project | GitHub
- Predicted U.S. power outage duration across 16 years of continental data (2000–2016), engineering geographic, climatic, and infrastructure features to achieve 80% accuracy with a tuned Random Forest.

Skills

Languages: Python, SQL (PostgreSQL), R, JavaScript, HTML, CSS, Java
Libraries: pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Streamlit, PySpark, Dask, PyTorch, LangChain, FinBERT, MiniLM, FAISS
Tools: Git, GitHub Actions, Docker, AWS (Lambda, Bedrock), DVC, pytest, Jupyter, Bash, Neo4j, Google Earth Engine
Core Competencies: Machine Learning, Deep Learning, NLP, RAG, Statistical Modeling, Time Series Analysis, EDA, Data Visualization